

3D scanning software

- Rapidform
- Easy3D Scan
- SolidWorks
- Geomagic
- Polyworks
- Delcam

3D modelling software

- AutoCAD
- Blender
- Pro Engineer
- CATIA
- Unigraphics
- SolidWorks
- trueSpace

3D modelling helps to check the assembly feasibility of that product with another part," Singh adds. With prototype building, the time consumed is less and the chances of errors are also minimised.

Reverse engineering for improvisation

Reverse engineering is the process of extracting the design from an existing product, with an aim to modify and improve it for the next version of the product, for value engineering or bug fixing. It is most commonly used in automotive, aerospace and consumer electronics industries. Also, "Reverse engineering is important to update products when there is little or no documentation available on the design of the product," shares Rawat.

Dr Vaidya explains, "Reverse engineering connotation can also be used when we are trying to mimic some product and replicate it in a new form. The vital principle while replicating the process is that it has to fall within the legal framework."

Reverse engineering is creating 2D and 3D drawings to regenerate computer-aided designs of an existing product. It is a tedious process to find out all the dimensions, and even if one dimension is missed, you would be unable to replicate the whole thing.



Concept design of a cellphone

You may have experienced this in some designs. For example, while talking on a touch-screen phone, your ear touching the touch screen may cancel the call. Diwan says, "Due to reverse engineering, there are fewer chances for failure. It helps to analyse good and bad features so that the desired features can be retained for a long term, while faulty/bad features can be designed out."

"Reverse engineering is also used in benchmarking competing products and assessing patent infringements. Given the stiff competition in the market today, it is becoming an important tool to stay ahead of times," adds Rawat.

Automated verification

Apart from modelling and reverse engineering, 3D laser scanning is used in quality assurance. Rawat says, "Scanners are placed at the end of production lines to capture the 3D data of pieces as they come off the line. The 3D scanned data is validated against the original CAD data to provide an automated verification of the

product. 3D laser scanning also has the capacity to acquire hundreds of thousands or millions of points spread across the entire geometrical figure of the scanned material and thus makes it possible to accurately describe the freeform surfaces and digitise the entire components that may lie inside the object."

"3D scanning helps in inspection and rectification. Often, production tooling fixtures undergo a deformation or damage after a certain time period. If the native CAD data is unavailable for such a fixture, it becomes hard to repair or replicate the same fixture. With 3D laser scanning, one can recreate a local insert which has been damaged or the entire fixture more accurately and much faster than any other 3D coordinate measuring system," adds Rawat.

3D measurements for concept design

It is hard to believe but cellphone companies spend a lot of time and money in creating superb concept phones